



Colorizing Image through Regression

- **Goal:**
Automatically colorize grayscale images into realistic RGB outputs
- **Dataset:**
<https://www.kaggle.com/datasets/akash2sharma/tiny-imagenet>
- **Models:**
Linear Regression / SVR
Neural Network (Baseline) / U-Net (Deep Learning)



Modeling Strategy

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- Convert images into:
 - L channel (grayscale)
 - a, b channels (color)
- Train models to predict color channels

Methods Compared

- Traditional ML: Linear Regression & SVR
- Deep Learning: Neural Network & U-Net

Performance comparison

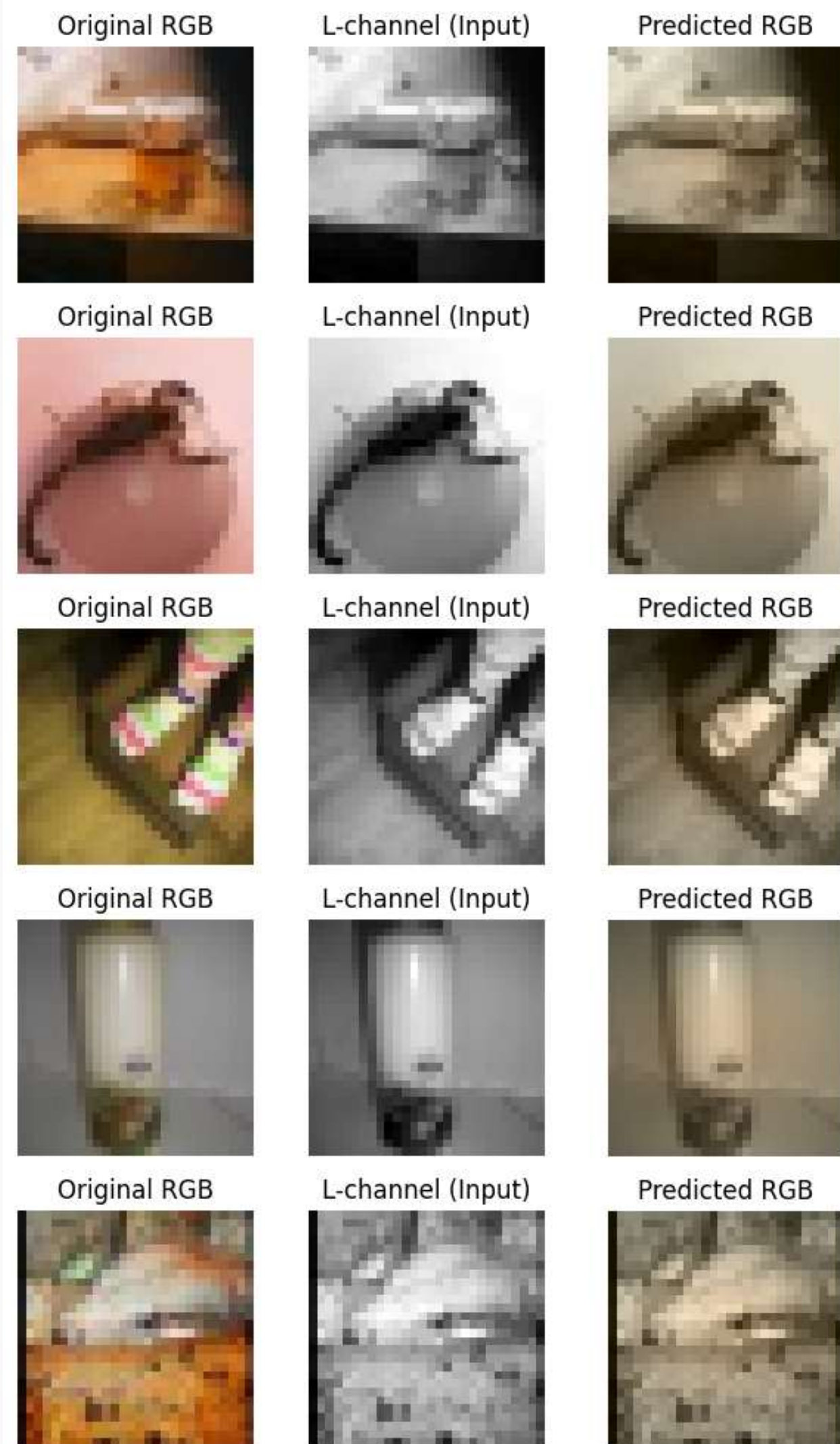
MSE / L1 Loss Measures pixel-level error between predicted and ground truth color values. This metric reflects overall difference, and lower values indicate better performance.	SSIM Structural Similarity Index — evaluates how well image structure is preserved. This metric reflects perceptual similarity, and higher values (closer to 1) indicate better alignment with the ground truth.	PSNR Peak Signal-to-Noise Ratio — measures overall image quality in decibels. Higher values indicate less noise and better reconstruction quality.
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Metrics	R^2	MSE	L1	SSIM	PNSR
Linear Regression	-0.0367	235.9506	-	0.9287	22.08 dB
SVR	-0.051	236.69	-	0.91	22.89 dB
Baseline (MSE Loss)	-	0.0114489	0.084472	0.6624	18.56 dB
Baseline (L1 Loss)	-	0.014991	0.084288	0.6619	18.40 dB
Baseline (Hybrid Loss)	-	0.014803	0.083720	0.6723	18.47 dB
U-Net (MSE Loss)	-	0.007858	0.055793	0.9320	21.51 dB
U-Net (L1 Loss)	-	0.055431	0.008083	0.9371	21.48 dB
U-Net (Hybrid Loss)	-	0.007893	0.054121	0.9387	21.59 dB



Linear Regression: Result

Metric	Value
R^2	-0.0367
MSE	235.9506
SSIM	0.9287
PSNR	22.08



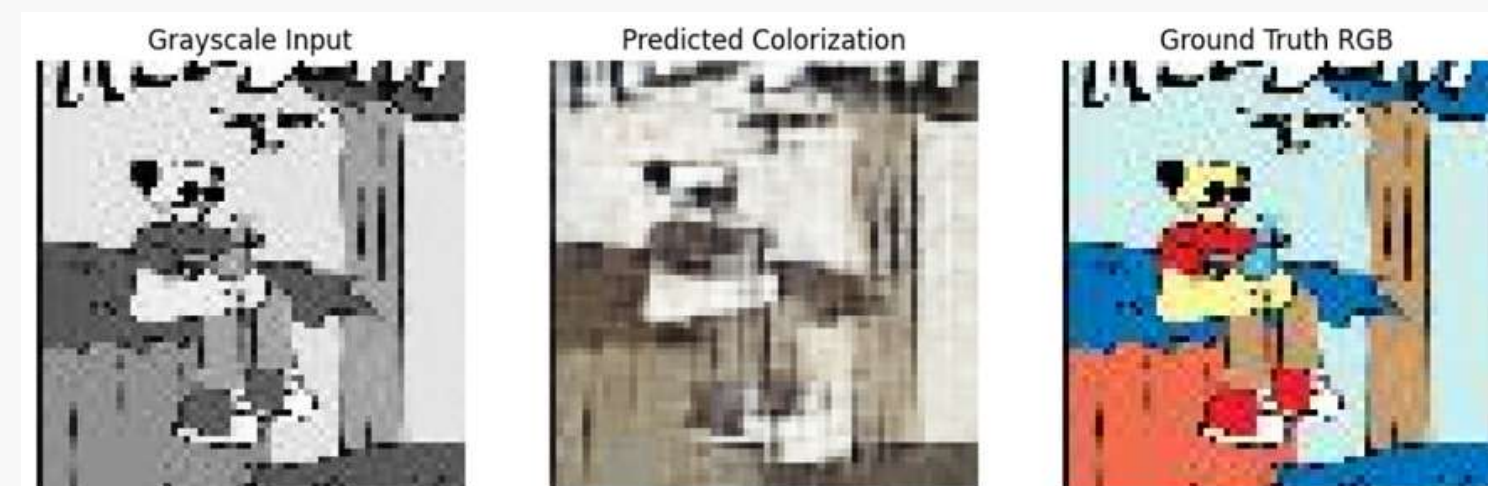
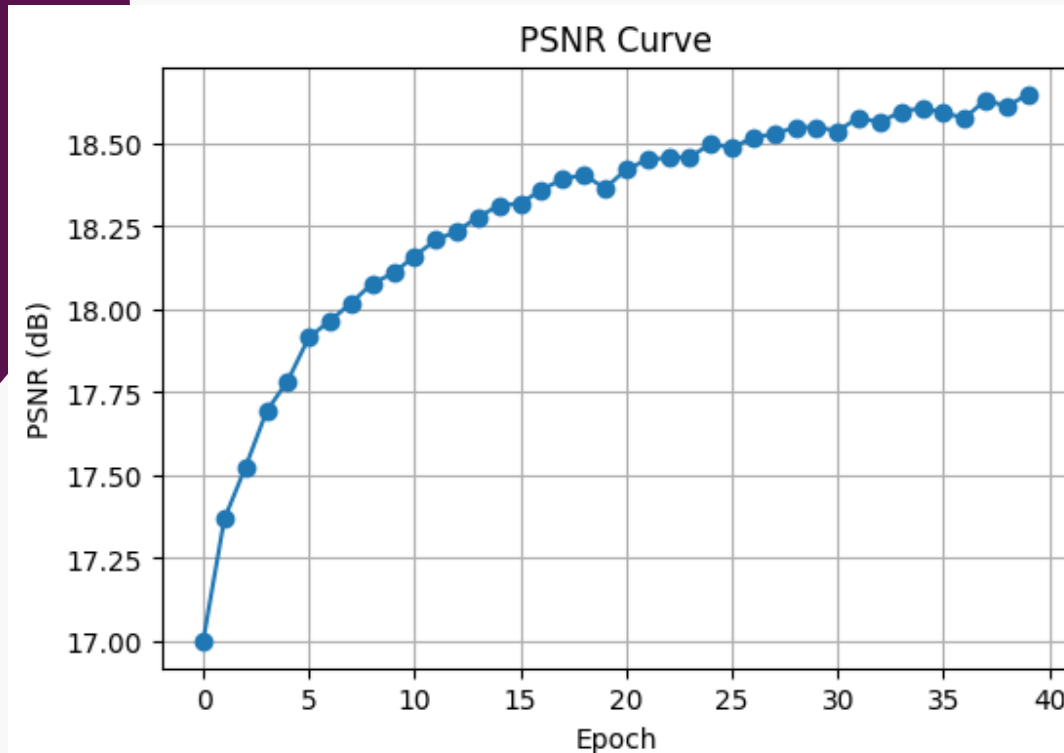
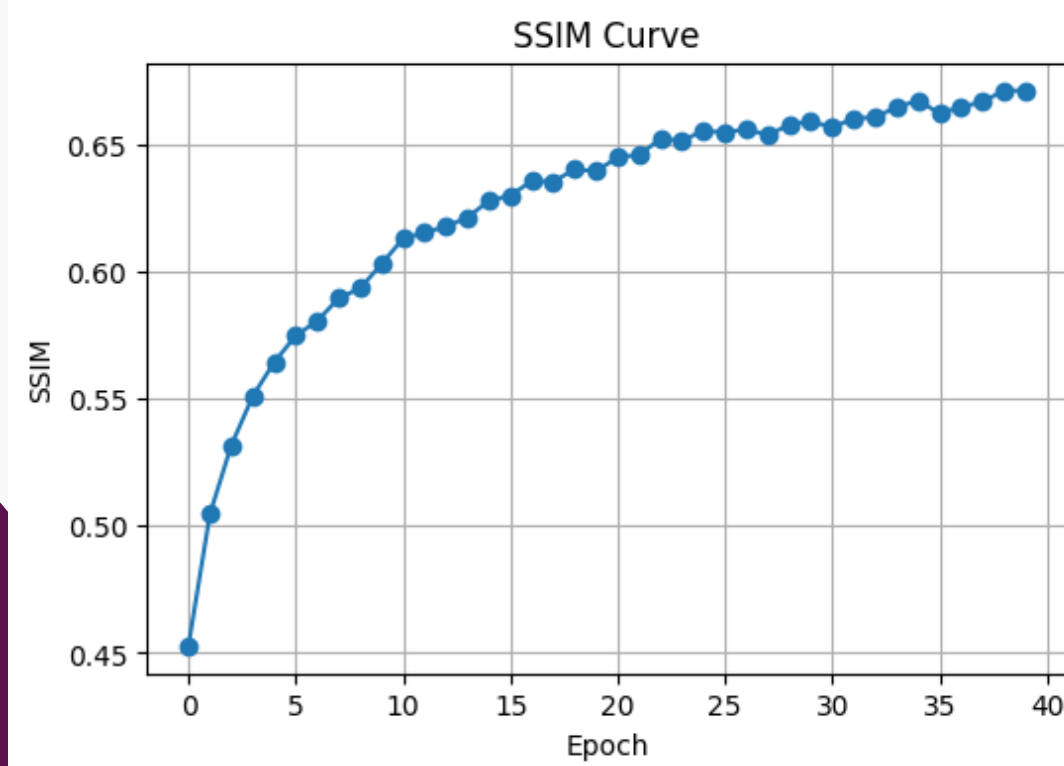
SVR: Result

Metric	Value
R^2	-0.051
MSE	236.69
Mean SSIM	0.91
Mean PSNR	22.89

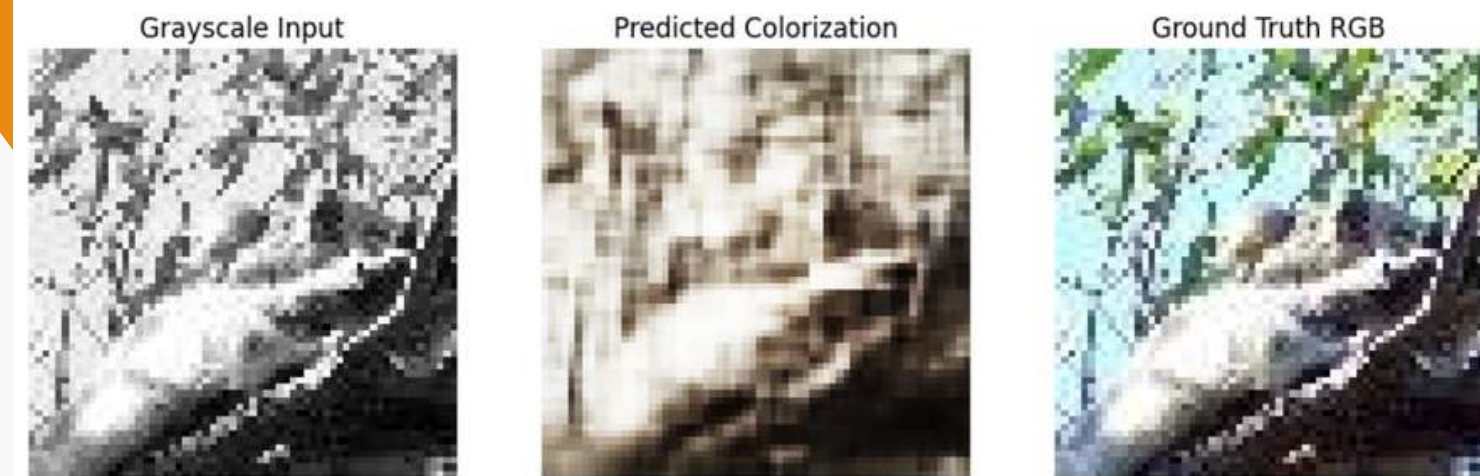
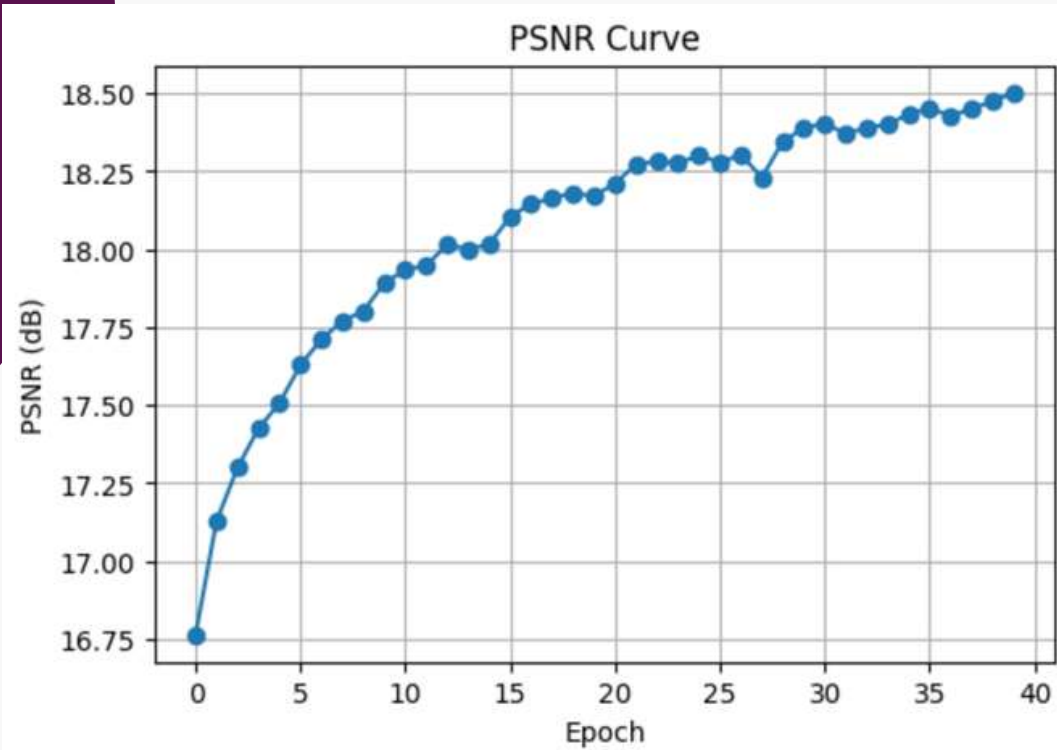
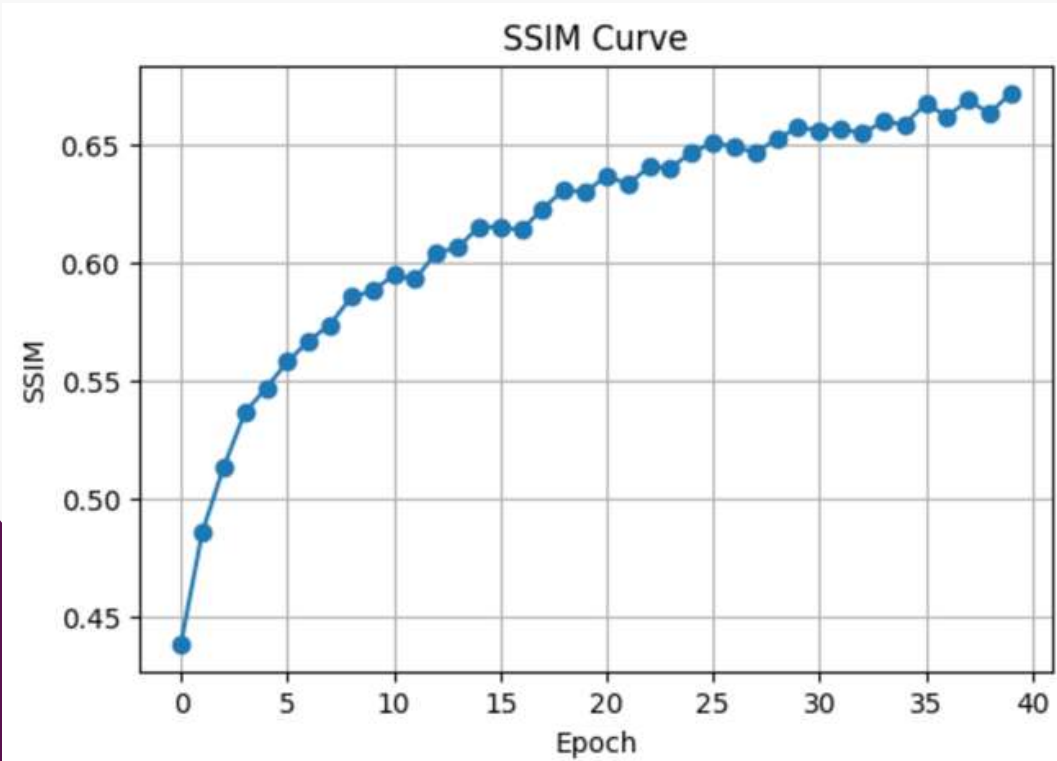
Disadvantage:

Computationally expensive and very slow

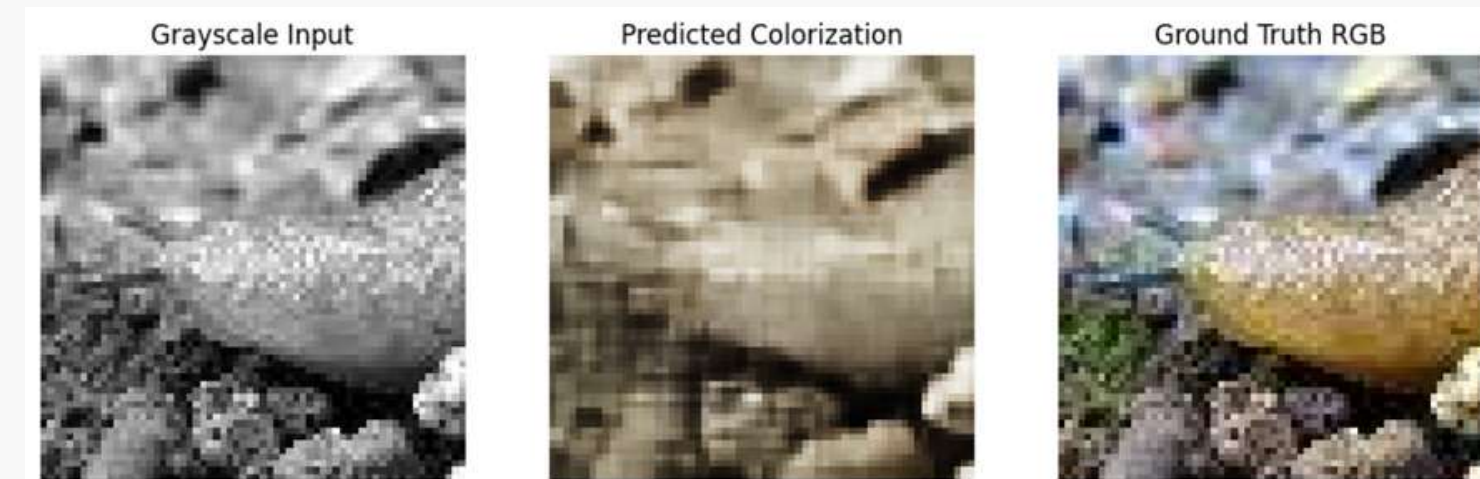
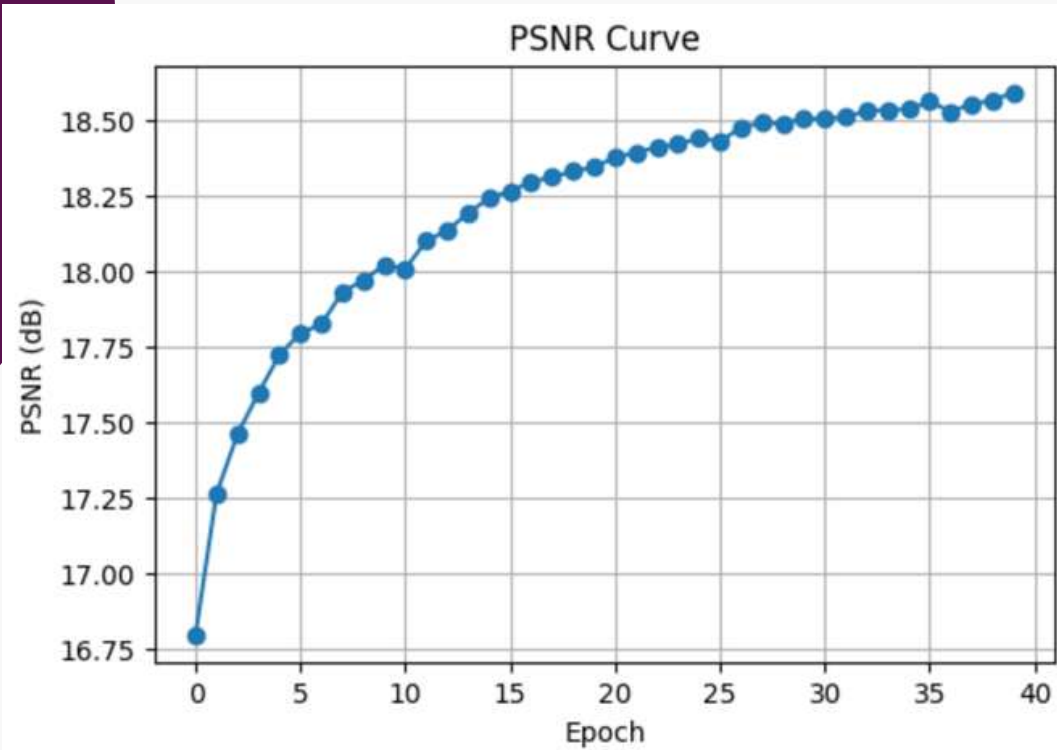
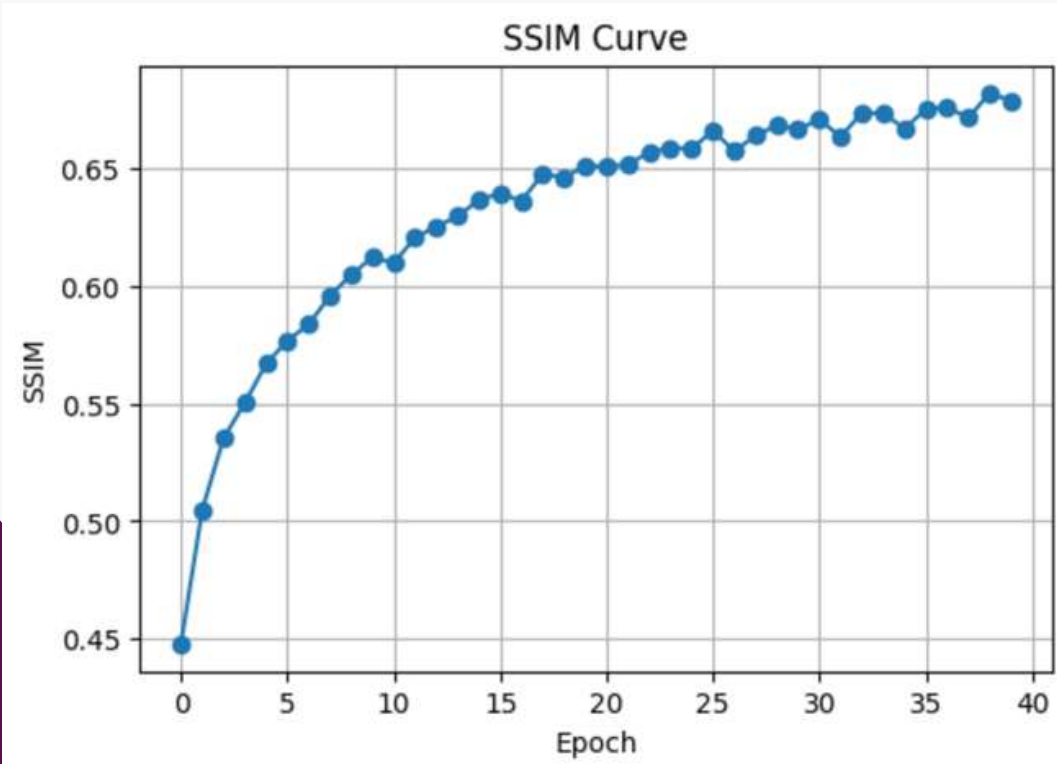
Baseline — MSE Loss



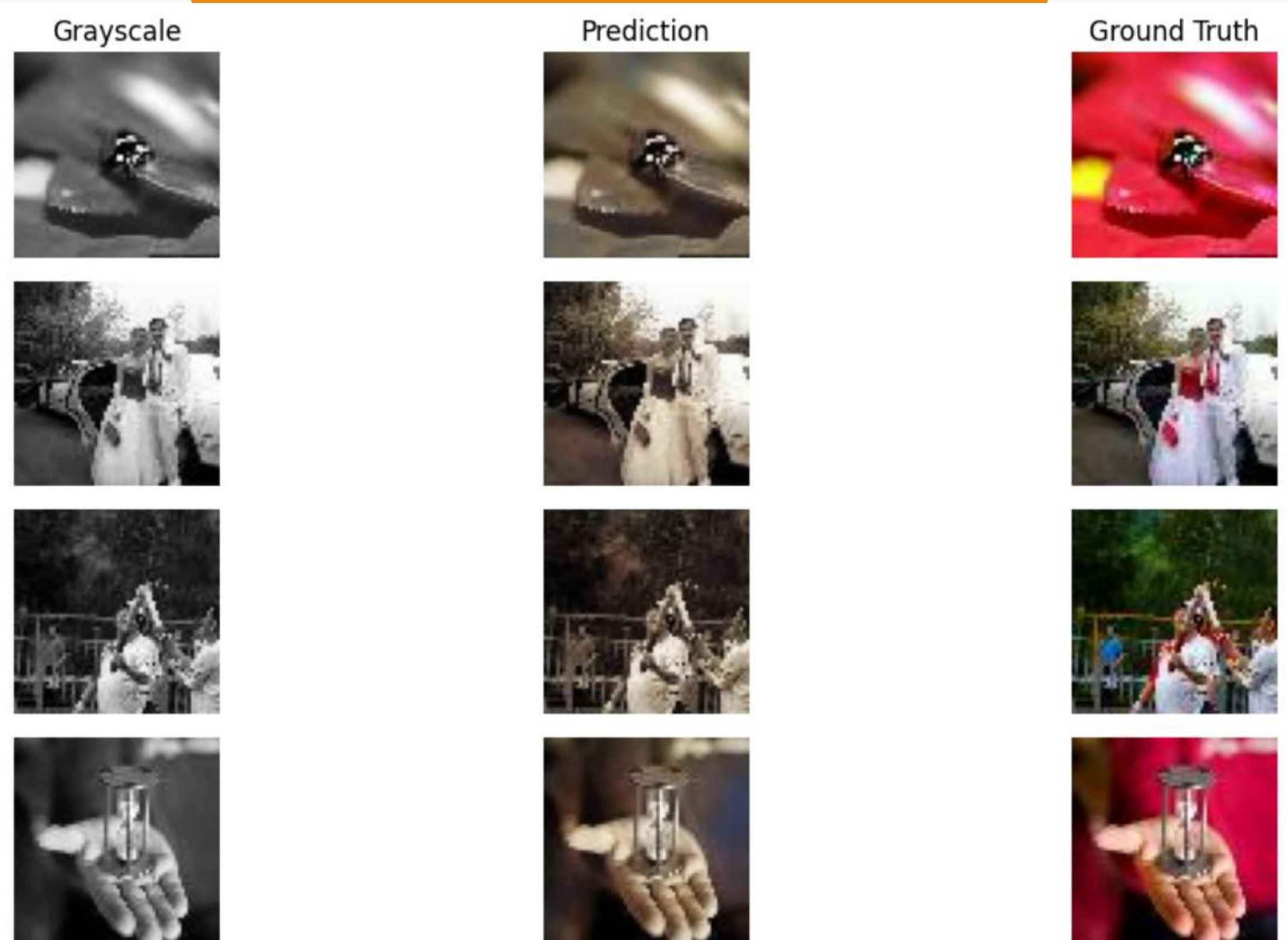
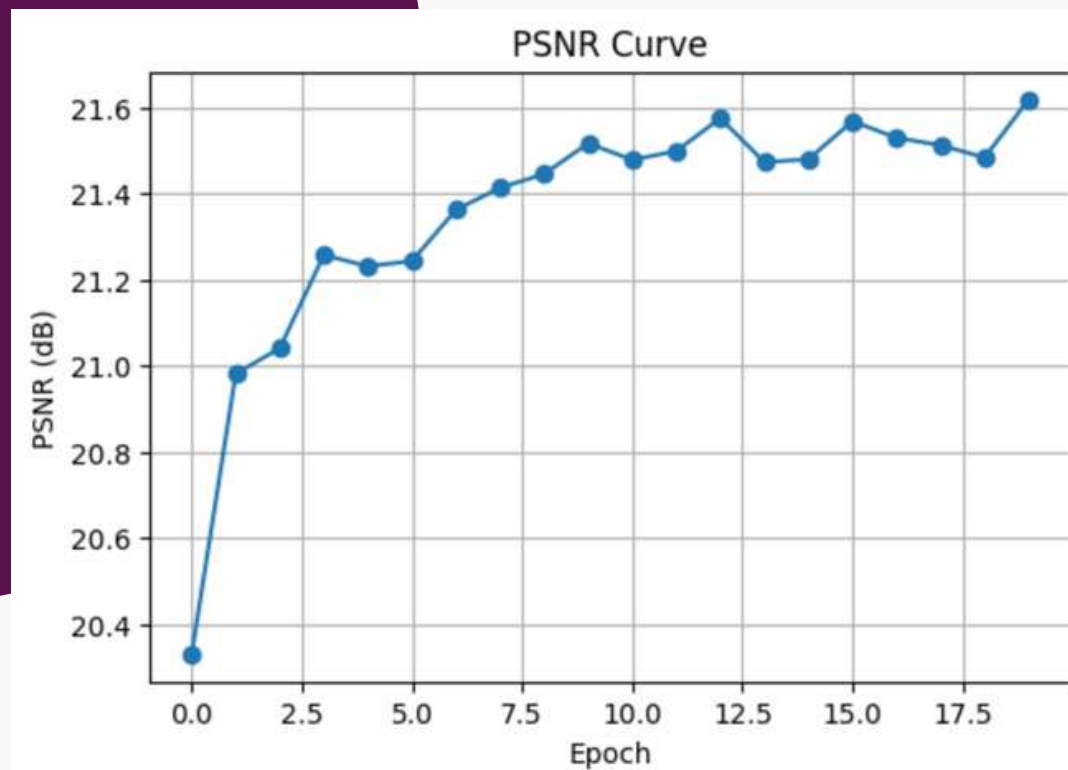
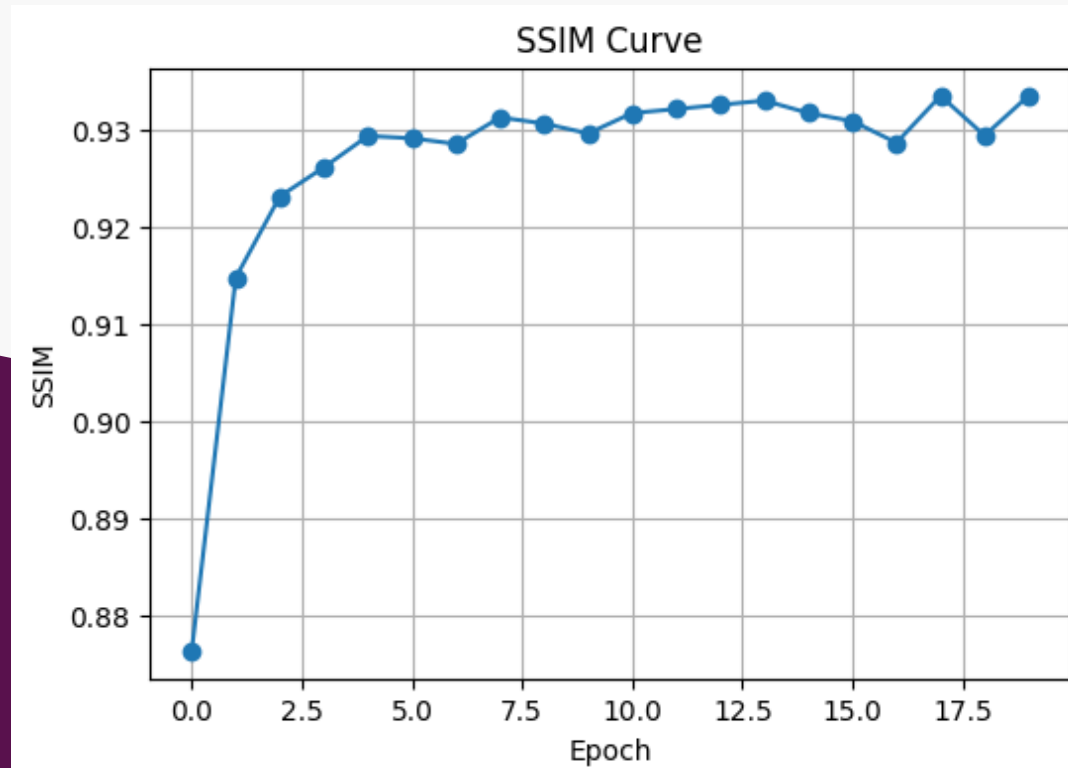
Baseline — L1 Loss



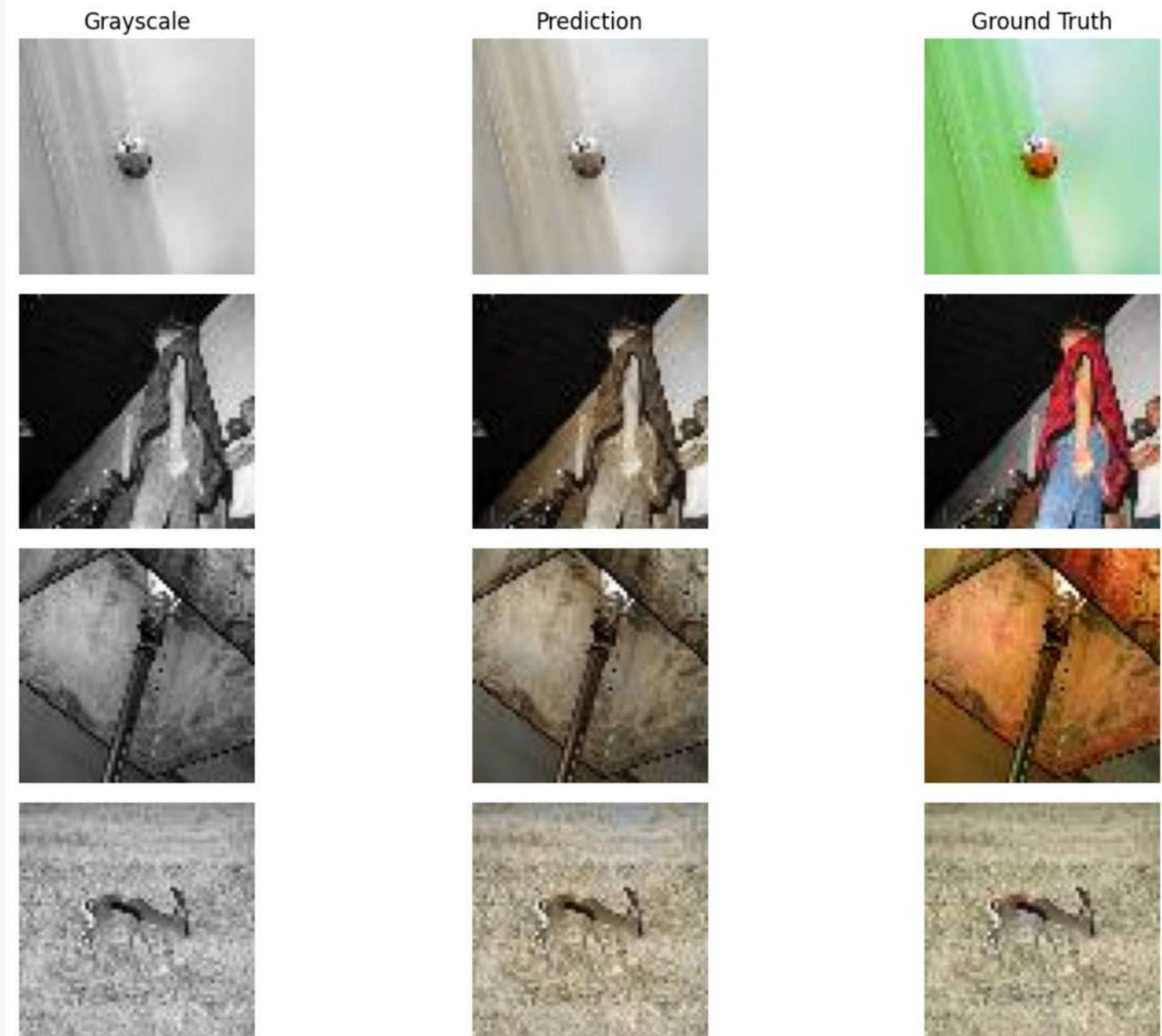
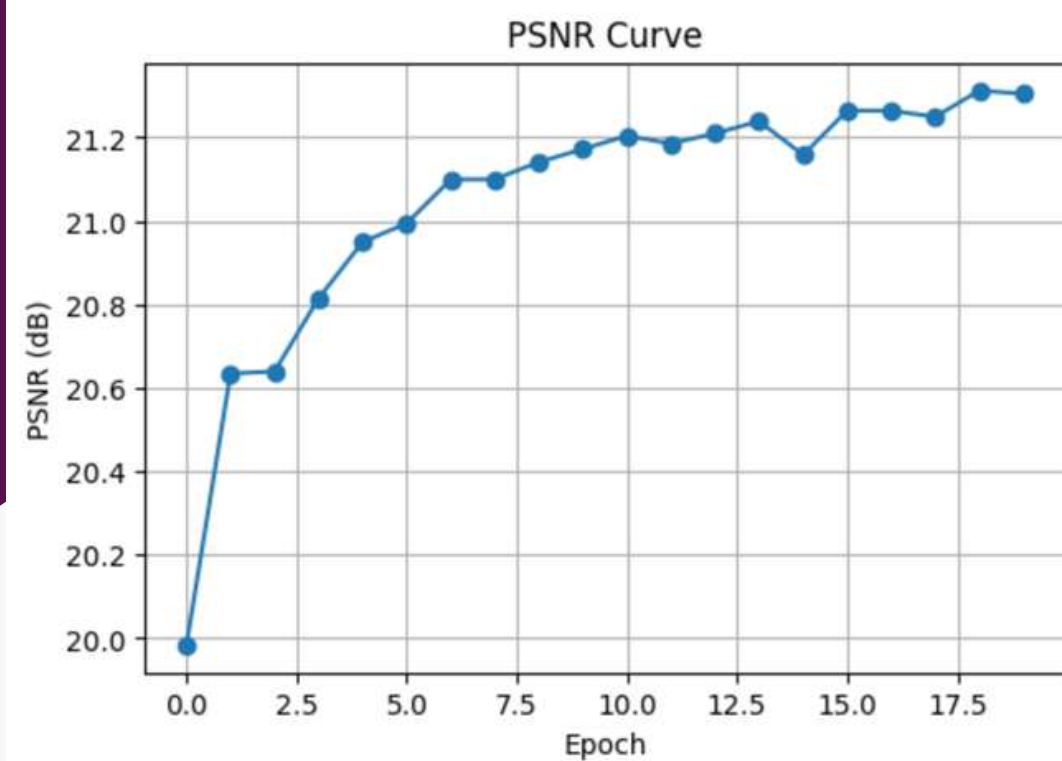
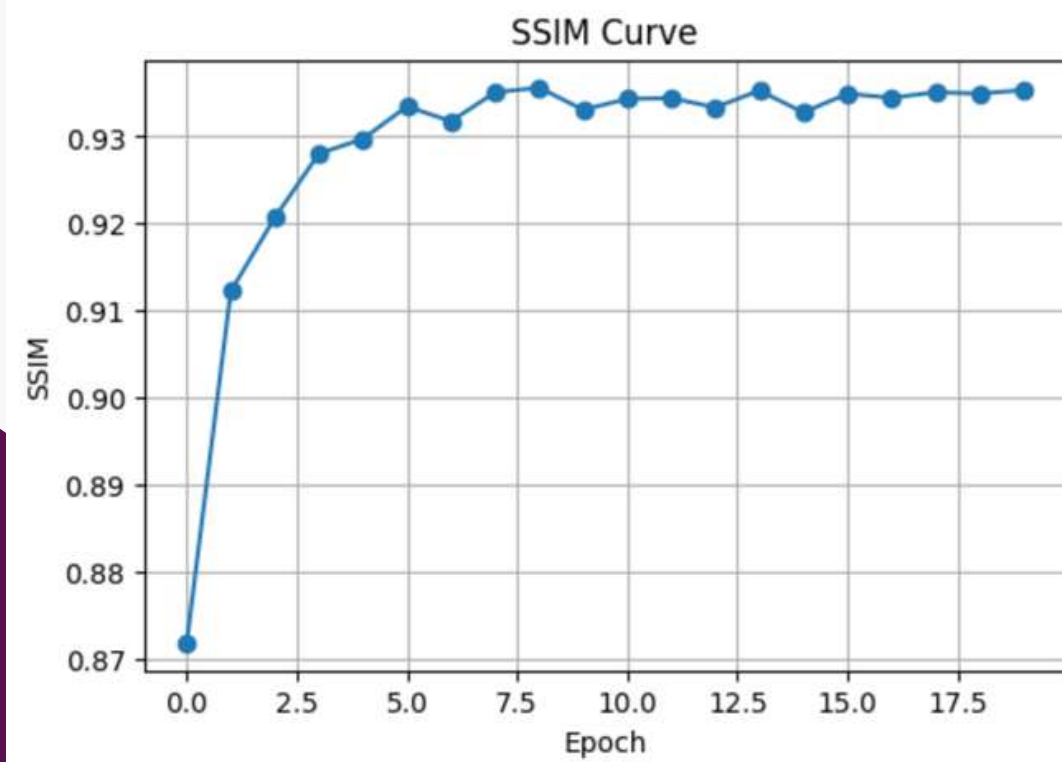
Baseline — Hybrid Loss



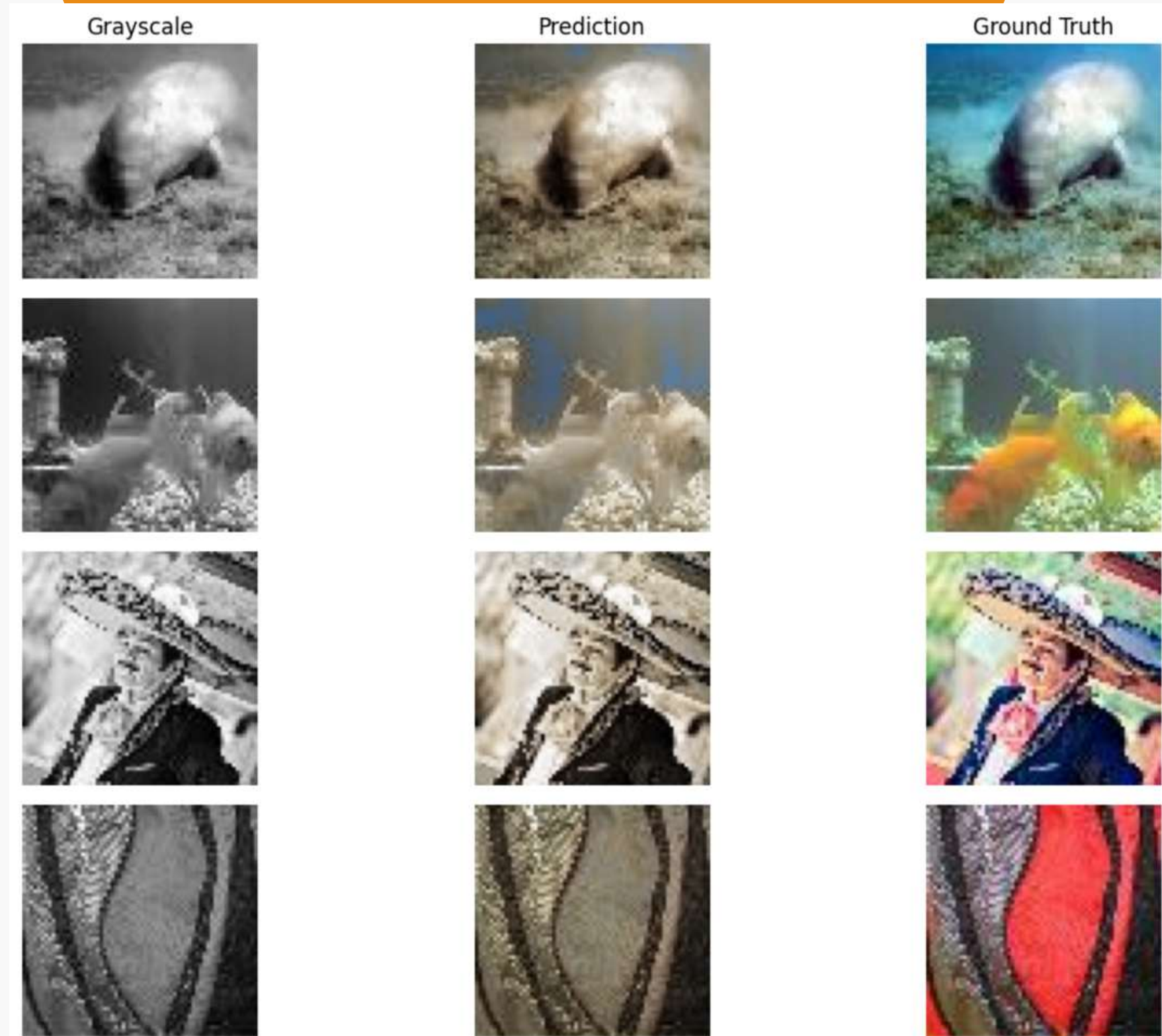
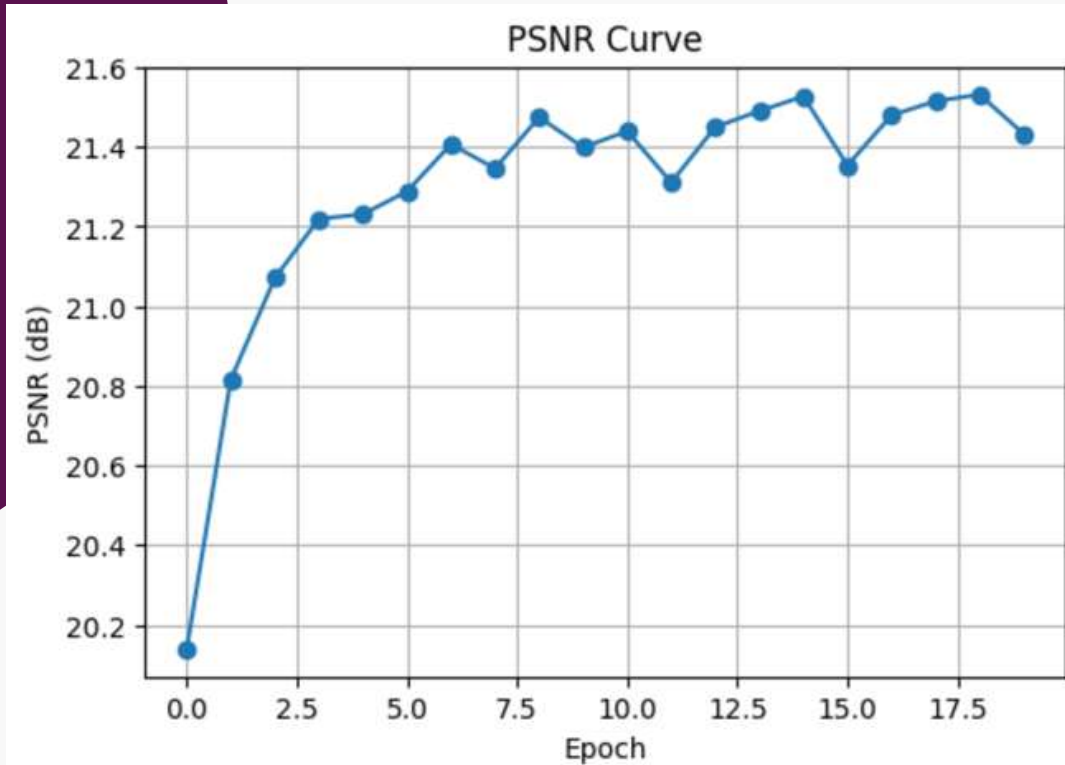
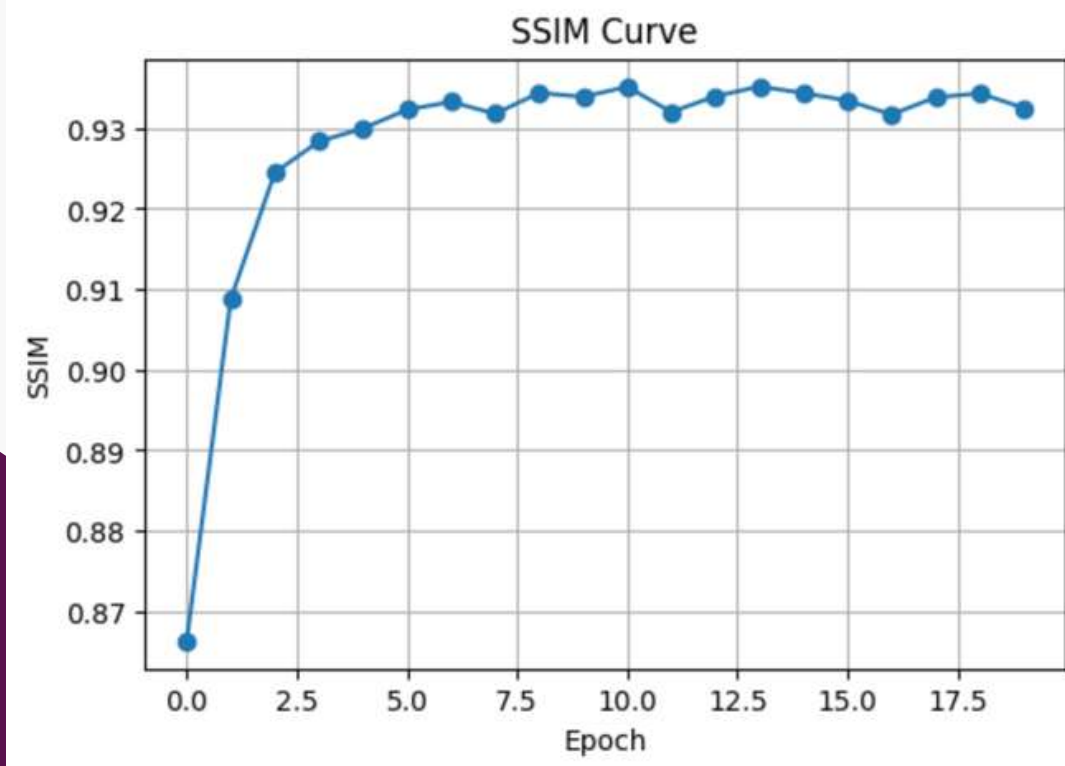
U-Net — MSE Loss



U-Net — L1 Loss



U-Net — Hybrid Loss





Key Insights

- Traditional models achieve higher PSNR due to smoother predictions, but fail to produce realistic colors.
 - U-Net, while having slightly lower PSNR, delivers significantly better perceptual quality by capturing spatial structure.
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Future Work

- Explore GAN-based colorization models (e.g., Pix2Pix, cGAN) for more realistic colors.
- Incorporate SSIM Loss into the training objective

$$\text{Hybrid Loss} = \alpha(L1) + \beta(MSE) + \gamma(1 - SSIM)$$

- Enhance the U-Net architecture with skip connections



Thank you

Please direct any further questions to presenter

Contact Me

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